

Divisibility & Digital Roots

Quick tests to check divisibility without dividing, and the digital-root shortcut that powers many of them.

The divisibility tests

Divisible by	Quick test
2	last digit is even (0, 2, 4, 6, 8)
5	last digit is 0 or 5
10	last digit is 0
4	the last two digits form a number divisible by 4
8	the last three digits form a number divisible by 8
3	the sum of the digits is divisible by 3
9	the sum of the digits is divisible by 9
6	divisible by both 2 and 3
11	the alternating sum of digits (from the units digit) is 0 or a multiple of 11

The digital root

Keep adding the digits of a number until a single digit remains — that is the **digital root**. It equals the remainder when the number is divided by 9 (and a multiple of 9 has digital root 9).

$$7309 \rightarrow 7 + 3 + 0 + 9 = 19 \rightarrow 1 + 9 = 10 \rightarrow 1 + 0 = 1. \text{ Digital root} = 1$$

Why digit sums work for 9: every place value (10, 100, 1000...) is one more than a multiple of 9, so each digit contributes itself to the remainder. That's why the digit sum tells you the remainder on division by 9 (and the same idea, with 3, works for 3).

Prerequisite refresher: "Alternating sum" means +, −, +, − the digits. For 11, start the + at the units digit.

Where marks slip away

Trap 1 — 9 is not about the last digit. Both 99 and 109 end in 9, but only 99 is divisible by 9. Use the digit sum.

Trap 2 — Divisible by 4 and 6 $\neq$ divisible by 24. Check 24 by testing 3 and 8 (its co-prime factors), not 4 and 6.

Trap 3 — The 11 test. A result of 0 also passes — 0 is a multiple of 11.

Trap 4 — Digit sum gives the remainder. If the digit sum of a number is 30, it is divisible by 3 (30 is) but not by 9 (30 isn't).

Slip 5 — For 6, both tests must pass — divisible by 2 AND by 3.

Model answers — write them like this

Example A. Is 405 divisible by 9?

Step 1: Add the digits: $4 + 0 + 5 = 9$.

Step 2: 9 is divisible by 9, so 405 is divisible by 9.

$\therefore$ Yes, 405 is divisible by 9

Example B. Find the value(s) of z so that 31 z 5 is a multiple of 9.

Step 1: Digit sum = $3 + 1 + z + 5 = 9 + z$.

Step 2: $9 + z$ must be a multiple of 9, so $z = 0$ or $z = 9$.

$\therefore z = 0$ or 9

Example C. Check whether 406839 is divisible by 3, 9 and 11.

Step 1: Digit sum = $4 + 0 + 6 + 8 + 3 + 9 = 30$. Divisible by 3 (yes), by 9 (no, 30 isn't).

Step 2: Alternating sum from units: $9 - 3 + 8 - 6 + 0 - 4 = 4$, not a multiple of 11.

$\therefore$ Divisible by 3 only (not by 9 or 11)

Example D. Find the digital root of 7309.

Step 1: $7 + 3 + 0 + 9 = 19$.

Step 2: $1 + 9 = 10$, then $1 + 0 = 1$.

$\therefore$ Digital root = 1

Next: try the [worksheet](#), then check the [answer key](#).